

Experiment 2

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Aim

To design a user interface that allows users to recall visual elements such as icons or image chunks and evaluate how chunking improves memory recall.

Introduction

Chunking is a cognitive psychology technique where information is grouped into smaller meaningful units, making it easier for the brain to process and remember. In user interface design, chunking helps organize visual information effectively, allowing users to recall elements more easily.

This experiment demonstrates a UI where users view grouped visual elements for a short duration and then attempt to recall them. The effectiveness of chunking is evaluated by comparing scores from an unchunked display versus a chunked display.

UI Design Screens

The interface consists of five main screens designed to simulate the memory recall experiment.

1. Instruction Screen

Memory Recall Task

You will be shown several groups of words.

You will have 5 seconds to view them.

After that, recall the items you remember.

START

Instruction Screen:

The instruction screen introduces the memory recall task to the user.

Features

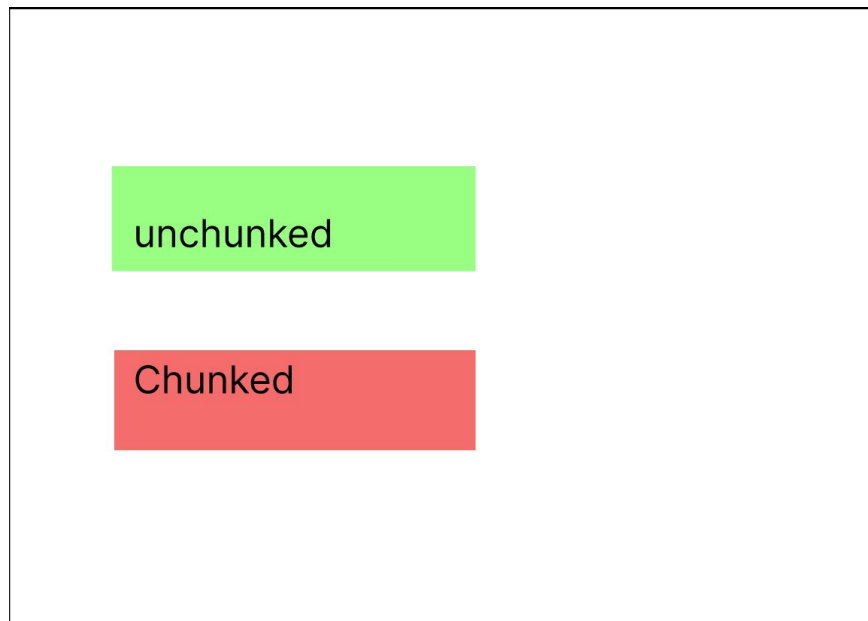
- Displays the title "**Memory Recall Task.**"

- Provides clear instructions explaining the memory task in three lines.
- Users are informed that they will be shown several groups of words and have **5 seconds** to view them.
- After viewing, they must recall the items they remember.
- A **START button** is provided to begin the experiment.

Purpose

This screen prepares the user and ensures they understand the task before proceeding.

2. Chunking Type Selection Screen



Type Selection Screen:

This screen allows the user to choose between the unchunked and chunked display modes.

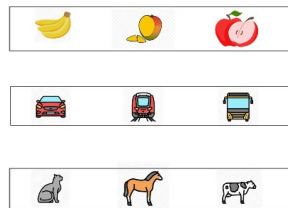
Features

- Presents two clearly labelled buttons: **unchunked** (green) and **Chunked** (red).
- Each button leads to a different display mode to compare memory recall performance.
- The color coding helps users visually differentiate the two modes.
- Simple, minimal layout reduces cognitive load before the task begins.

Purpose

This screen lets users experience both chunked and unchunked presentations and compare their recall effectiveness.

3. Chunked Display Screen



chunked

Chunked Display Screen:

This screen displays visual elements grouped into meaningful categories.

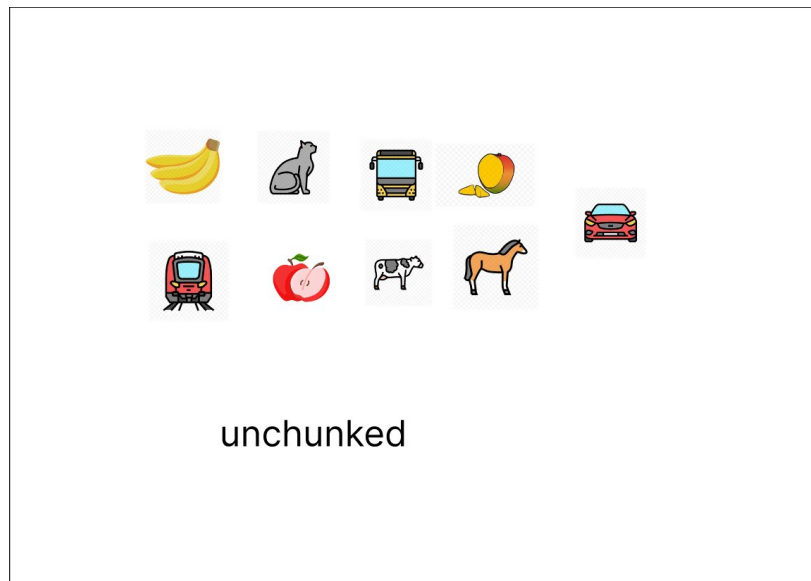
Features

- Icons are arranged into **three distinct grouped rows (chunks)**.
- Each group contains related icons such as:
 - Fruits (Banana, Mango, Apple)
 - Vehicles (Car, Train, Bus)
 - Animals (Cat, Horse, Cow)
- Icons within each group are placed inside a visible bounding box to clearly represent chunks.
- Labelled as "**chunked**" at the bottom for user awareness.

Purpose

This screen demonstrates how chunking organizes information into meaningful groups to help users remember visual elements more effectively.

4. Unchunked Display Screen



Unchunked Display Screen:

This screen displays the same set of visual elements in a randomized, ungrouped layout.

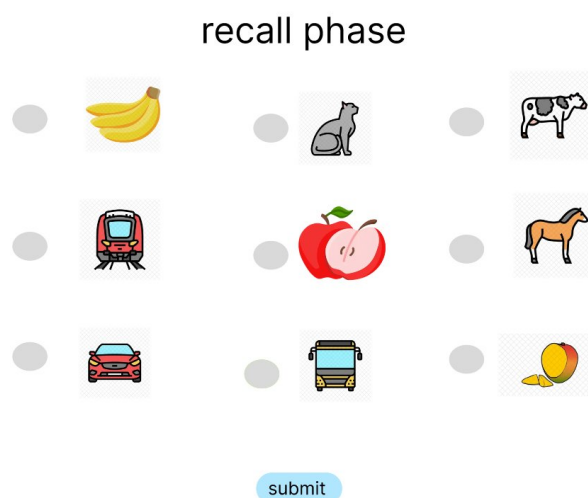
Features

- All 9 icons are scattered in a **random, ungrouped grid** layout.
- Items from different categories (fruits, vehicles, animals) are mixed together.
- No visual grouping or bounding boxes are provided to aid memory.
- Labelled as "**unchunked**" at the bottom to distinguish it from the chunked version.

Purpose

This screen serves as the control condition to compare against chunked recall, demonstrating that ungrouped information is harder to remember.

5. Recall Screen



Recall Screen:

The recall screen allows users to identify the items they remember from the display phase.

Features

- Displays a 3×3 grid of icons with radio buttons next to each item.
- Users must select (check) the icons they remember seeing.
- Icons used in recall include: banana, cat, cow, train, apple, horse, car, bus, mango.
- A **submit button** is provided to finalize the user's recall choices.

Purpose

This screen measures the user's ability to recall visual elements after viewing chunked or unchunked information.

6. Score / Result Screen

YOU SCORED 7 POINTS
OUT OF 10

Score Screen:

The score screen displays the user's performance after submitting their recall.

Features

- Shows the total score (for example **7 points out of 10 items recalled**).
- Score is displayed in large, bold uppercase text for clear readability.
- Minimal, clean layout so the result is the sole focus of the screen.
- Allows users to reflect on their recall performance before trying the other mode.

Purpose

This screen provides immediate feedback and allows users to evaluate and compare their memory performance across chunked and unchunked conditions.

Advantages of Chunking in UI Design

- Improves **memory retention**
- Makes information **easier to process**
- Reduces **cognitive load**
- Enhances **user experience and usability**
- Helps users **recall information faster**

Result

The UI design demonstrates how chunking helps users recall visual information more effectively. By grouping related icons into categories such as **Fruits, Vehicles, and Animals**, users can remember more items compared to the unstructured unchunked layout. The score screen confirms this by showing improved recall when items are presented in chunked form.